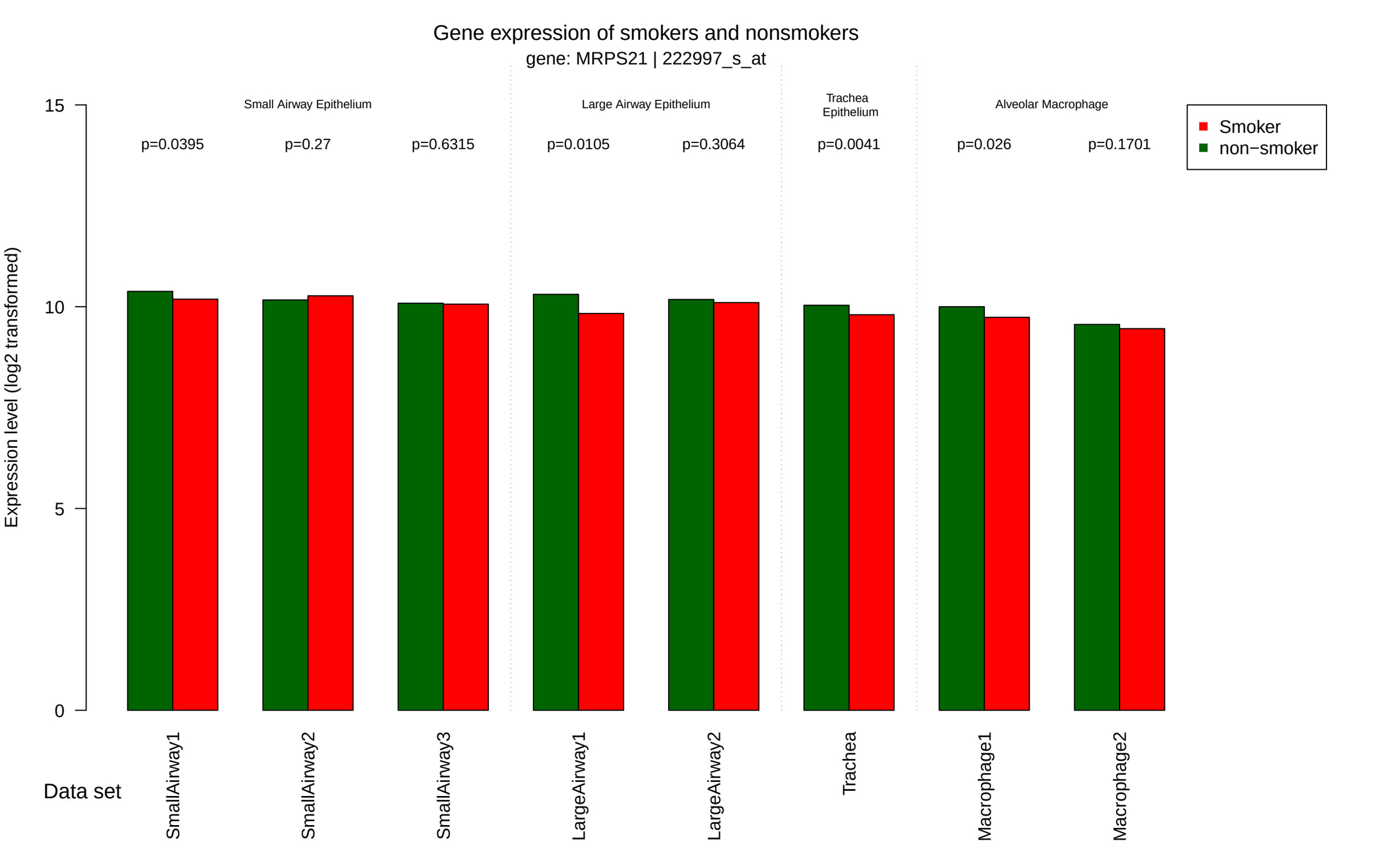
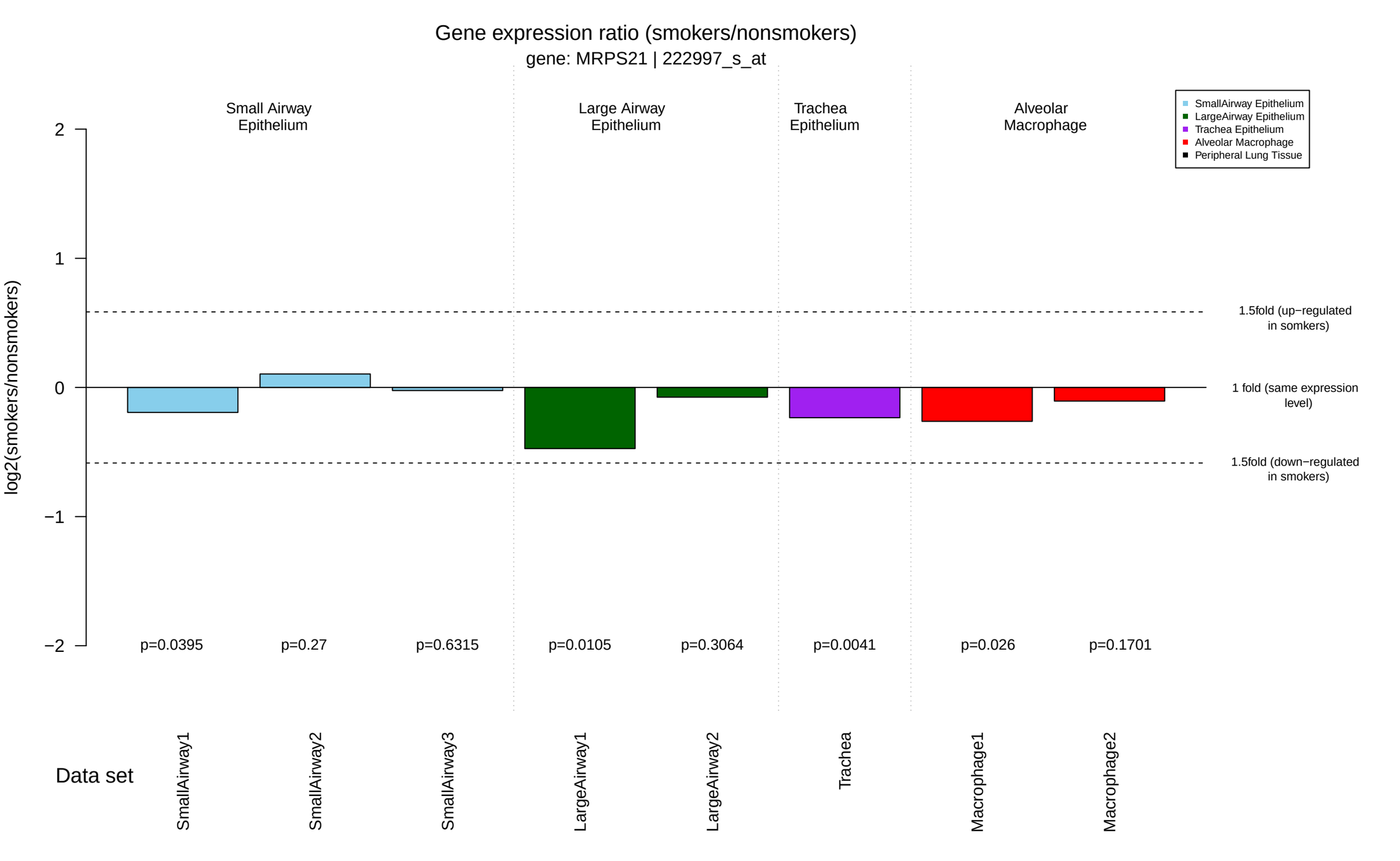
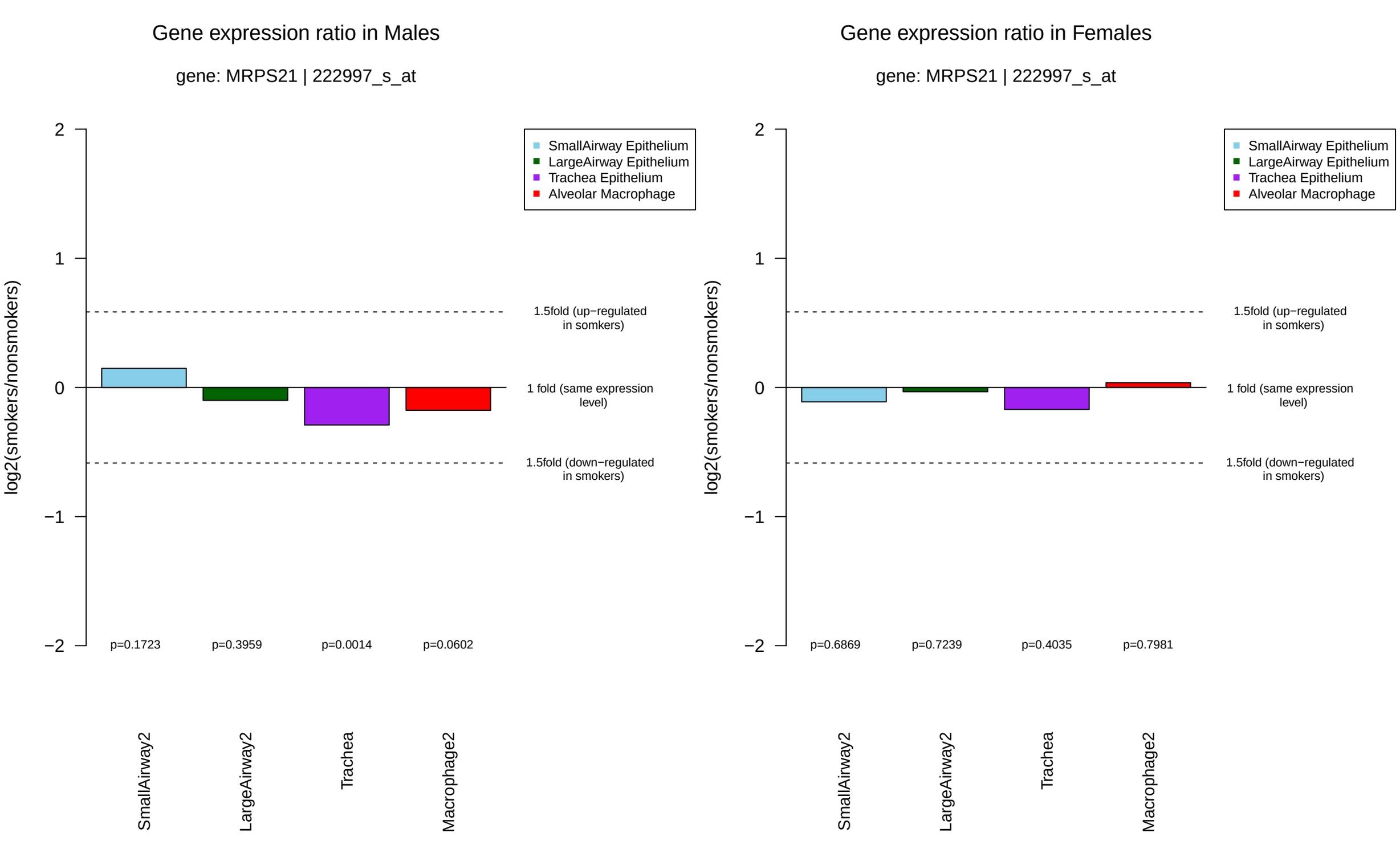
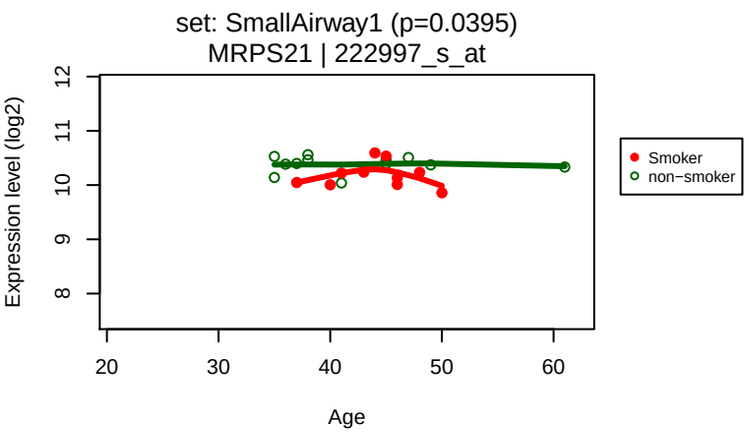
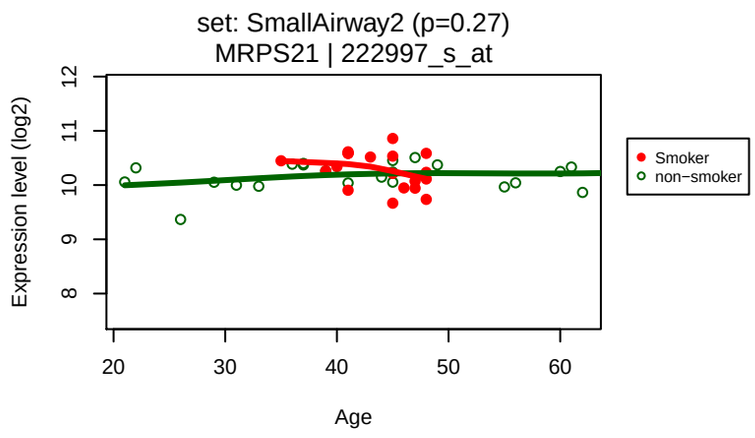
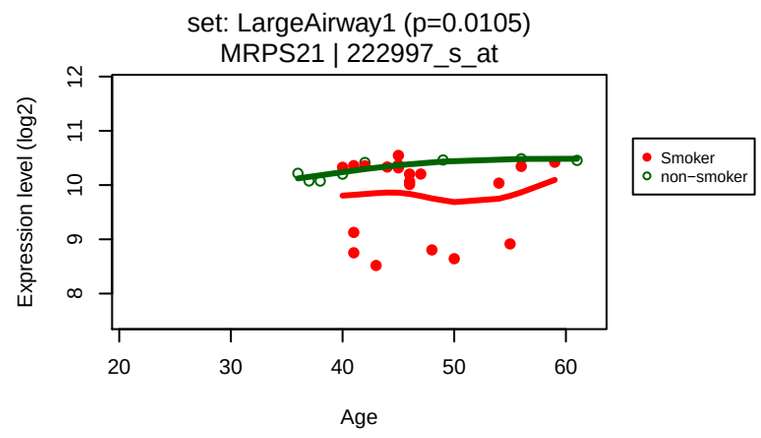

Gene expression of smokers and nonsmokers
gene: MRPS21 | 222997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0395
p=0.27
p=0.6315
p=0.0105
p=0.3064
p=0.0041
p=0.026
p=0.1701
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

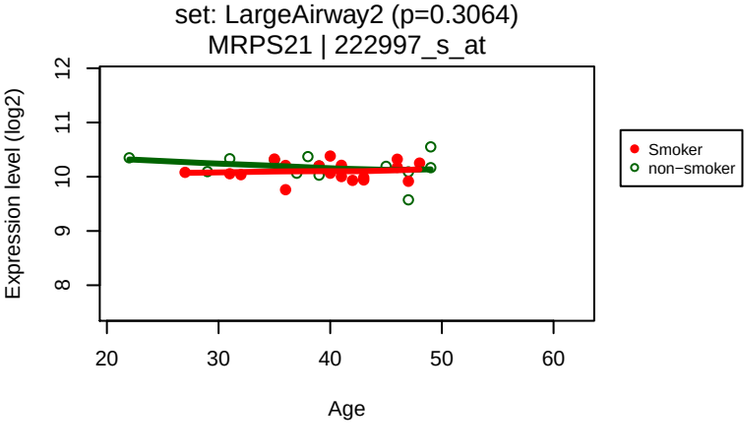

Gene expression ratio (smokers/nonsmokers)
gene: MRPS21 | 222997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0395
p=0.27
p=0.6315
p=0.0105
p=0.3064
p=0.0041
p=0.026
p=0.1701
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

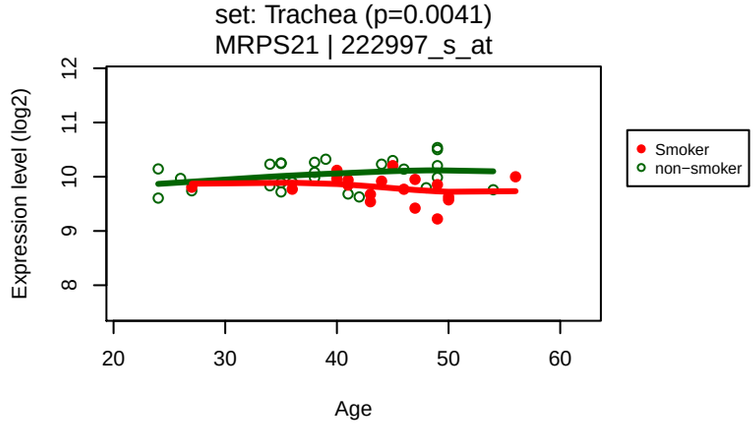

Gene expression ratio in Males
gene: MRPS21 | 222997_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.3959
p=0.0014
p=0.0602
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS21 | 222997_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6869
p=0.7239
p=0.4035
p=0.7981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

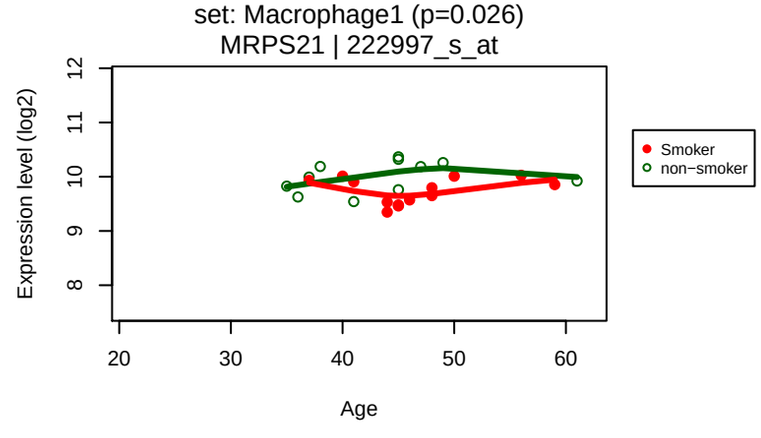

set: SmallAirway1 (p=0.0395)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker

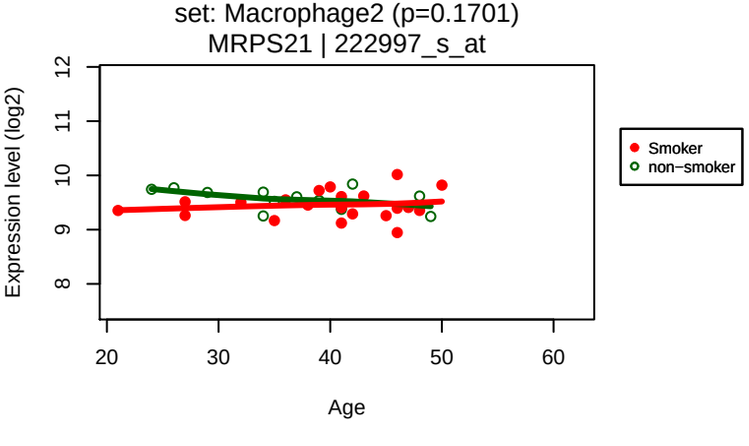

set: SmallAirway2 (p=0.27)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker

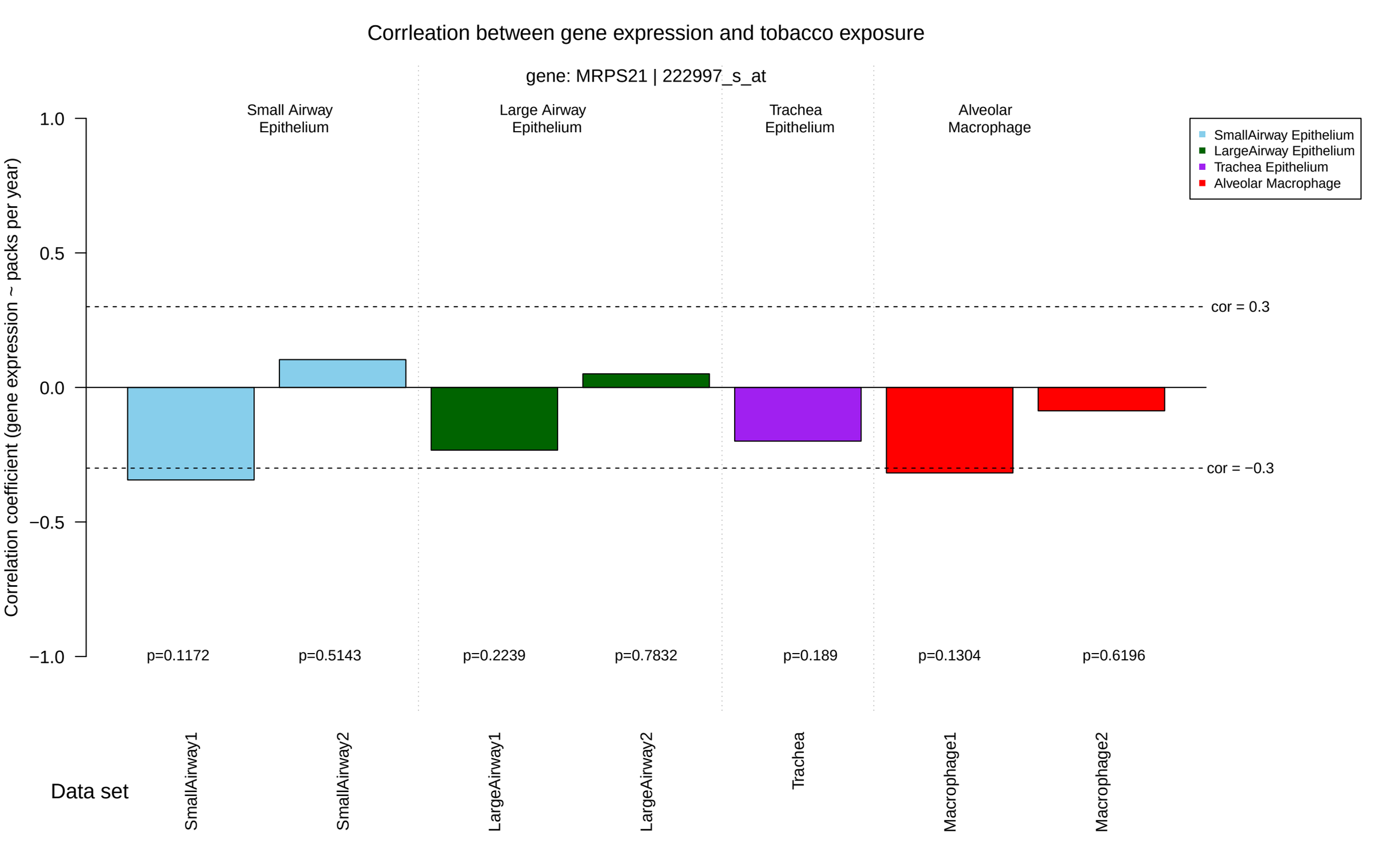

set: LargeAirway1 (p=0.0105)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3064)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0041)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.026)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1701)
MRPS21 | 222997_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS21 | 222997_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1172
p=0.5143
p=0.2239
p=0.7832
p=0.189
p=0.1304
p=0.6196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2